Analytical Chemistry Cumulative Exam.
January 26, 2018

1) For each of the following scientists, write a focused paragraph about what type of research they perform and what impact it has for the scientific community. Be sure to include major topic areas common to analytical chemistry and how these individuals have pushed the boundaries of these topic areas.

David Hage

Amy Herr

Amanda Hummon

2) Modern affinity chromatography uses monolithic supports. What is a monolithic support?

3) In a recent review article by David Hage and colleagues, they describe typical binding strengths for different types of affinity chromatography systems. In some cases, $K_{(association)}$ values are in the 10^6 M^{-1} or greater range and others 10^4 to 10^5 M^{-1} range. Explain why these values are important to know when designing an affinity chromatography system and how these values affect the separation. What type of separation phenomenon would be expected for the first type (10^6 M^{-1} or greater) vs. the 2nd type (10^4 to 10^5 M^{-1} range)?

4) What is a Western blot? Explain the physico-chemical processes involved in a Western blot.

5) Briefly explain what a Lab-on-a-Chip is (or micro-total analysis system, μ-TAS) and why there is significant interest in these devices for analytical chemistry. What are some of the physical processes used to make a lab-on-a-chip? Explain how these processes work from a fundamental level.

6) What is the transcriptome? Why is it challenging to analyze? Compare and contrast common methods to analyze the transcriptome.

7) What is the proteome? Why is it challenging to analyze? Compare and contrast common methods to analyze the proteome.